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Instructions

1. Given the matrices $A = \begin{pmatrix} 3 & 1 & 4 \\ 2 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, $B = \begin{pmatrix} 9 & 2 & 3 \\ 1 & 4 & 5 \\ 6 & 7 & 8 \end{pmatrix}$ 1 / 1 point

What is A+B?

- $\begin{pmatrix} 12 & 3 & 7 \\ 3 & 9 & 11 \\ 8 & 9 & 10 \end{pmatrix}$
- $\begin{pmatrix} 10 & 3 & 6 \\ 3 & 6 & 11 \\ 13 & 15 & 15 \end{pmatrix}$
- **(selected)** $\begin{pmatrix} 12 & 3 & 7 \\ 3 & 9 & 11 \\ 13 & 15 & 17 \end{pmatrix}$
- $\begin{pmatrix} 9 & 3 & 4 \\ 4 & 9 & 11 \\ 13 & 15 & 17 \end{pmatrix}$

Nice work

Correct! To add two matrices, you add their corresponding elements. For example, the element in the first row and first column of A (which is 3) is added to the element in the first row and first column of B (which is 9), giving 12 in the result matrix. Continue this for all corresponding elements to verify that the resulting matrix is:

$$A + B = \begin{pmatrix} 12 & 3 & 7 \\ 3 & 9 & 11 \\ 13 & 15 & 17 \end{pmatrix}$$

2. Given the matrices $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$ 1 / 1 point

What is A×B?

- $\begin{pmatrix} 4 & 4 \\ 10 & 6 \end{pmatrix}$
- **(selected)** $\begin{pmatrix} 4 & 4 \\ 10 & 8 \end{pmatrix}$
- $\begin{pmatrix} 5 & 4 \\ 11 & 10 \end{pmatrix}$
- $\begin{pmatrix} 8 & 4 \\ 14 & 8 \end{pmatrix}$

Nice work

Correct! To multiply two matrices, each element in the resulting matrix is obtained by taking the dot product of the corresponding row from matrix A and column from matrix B.

3. Given the matrix $A = \begin{pmatrix} 2 & 0 & -1 \\ 3 & 5 & 1 \\ 4 & -2 & 6 \end{pmatrix}$ 1 / 1 point

What is A^T (the transpose of A)?

- $\begin{pmatrix} 2 & 3 & 4 \\ -1 & 1 & 6 \\ 0 & 5 & -2 \end{pmatrix}$
- $\begin{pmatrix} 2 & 0 & -1 \\ 3 & 5 & 1 \\ 4 & -2 & 6 \end{pmatrix}$
- **(selected)** $\begin{pmatrix} 2 & 3 & 4 \\ 0 & 5 & -2 \\ -1 & 1 & 6 \end{pmatrix}$
- $\begin{pmatrix} -1 & 0 & 2 \\ 1 & 5 & 3 \\ 6 & -2 & 4 \end{pmatrix}$

Nice work

Correct! The transpose of a matrix is created by switching rows with columns.

4. Given the matrix $A = \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix}$, 1 / 1 point

What is the determinant of A, denoted as det(A)?

- 2
- 0
- **(selected)** -2
- 6

Nice work

Correct! Nice work on finding the determinant of the matrix.

5. In image processing, matrices are used to manipulate graphics. What does multiplying by the identity matrix I achieve in this context? 1 / 1 point

- It inverts the colors of an image.
- **(selected)** It preserves the image exactly as it is.
- It rotates the image by 90 degrees.
- It scales the image by a factor of 2.

Nice work

Great job! You're absolutely right. Multiplying by the identity matrix preserves the original data or image, as it acts as a neutral element in matrix transformations. This is a key concept in linear algebra, where the identity matrix plays the role of maintaining the original structure.

6. Imagine you have two matrices, $A(2 \times 3)$ and $B(3 \times 4)$. Which statement is **TRUE**? 1 / 1 point

- Multiplying $A \times B$ and $B \times A$ will yield matrices of the same dimensions.
- Both $A \times B$ and $B \times A$ are possible and result in square matrices.
- **(selected)** $A \times B$ is defined, but $B \times A$ is **not** defined.
- Matrix multiplication is commutative, so $A \times B = B \times A$

Nice work

Well done! You're absolutely right. Matrix multiplication $A \times B$ is possible because the number of columns in A matches the number of rows in B, resulting in a 2×4 matrix. However, $B \times A$ is not defined since the number of columns in $B(4)$ does not match the number of rows in $A(2)$. Understanding the dimensions is key to correctly applying matrix multiplication rules